

Converting Text into vectors with TF-IDF

Let's take an example where we have a corpus (a collection of documents) with three documents and our goal is to calculate the TF-IDF score for specific terms in these documents.

1. **Document 1:** "The cat sat on the mat."
2. **Document 2:** "The dog played in the park."
3. **Document 3:** "Cats and dogs are great pets."

Step 1: Calculate Term Frequency (TF)

For Document 1:

- The word "cat" appears 1 time.
- The total number of terms in Document 1 is 6 ("the", "cat", "sat", "on", "the", "mat").
- So, $TF(cat, Document\ 1) = 1/6$

For Document 2:

- The word "cat" does not appear.
- So, $TF(cat, Document\ 2) = 0$.

For Document 3:

- The word "cat" appears 1 time.
- The total number of terms in Document 3 is 6 ("cats", "and", "dogs", "are", "great", "pets").
- So $TF(cat, Document\ 3) = 1/6$

In Document 1 and Document 3 the word "cat" has the same TF score. This means it appears with the same relative frequency in both documents. In Document 2 the TF score is 0 because the word "cat" does not appear.

Step 2: Calculate Inverse Document Frequency (IDF)

- **Total number of documents in the corpus (D):** 3
- **Number of documents containing the term "cat":** 2 (Document 1 and Document 3).

$$IDF(cat, D) = \log \frac{3}{2} \approx 0.176$$

Step 3: Calculate TF-IDF

The TF-IDF score for "cat" is 0.029 in Document 1 and Document 3 and 0 in Document 2 that reflects both the frequency of the term in the document (TF) and its rarity across the corpus (IDF).

The TF-IDF score is the product of TF and IDF:

TF-IDF

- For Document 1: $\text{TF-IDF}(\text{cat}, \text{Document 1}, D) = 0.167 \times 0.176 = 0.029$
- For Document 2: $\text{TF-IDF}(\text{cat}, \text{Document 2}, D) = 0 \times 0.176 = 0$
- For Document 3: $\text{TF-IDF}(\text{cat}, \text{Document 3}, D) = 0.167 \times 0.176 \sim 0.029$